NAME : SANDESH NAKADE

CLASS : CSAI-D

ROLL NO : 34

PRN : 12412601

PROBLEM STATEMENT :Use the diabetes data set from UCI and Pima Indians Diabetes data set for performing the following: a. Univariate analysis: Frequency, Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis b. Bivariate analysis: Linear and logistic regression modeling c. Multiple Regression analysis d. Also compare the results of the above analysis for the two data sets.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
from sklearn.datasets import load_diabetes
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, confusion_matrix
```

```
df = pd.read_csv("heart_disease_dataset.csv")
```

```
df.head()
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	39	1	1	102	123	0	1	194	0	0.31	1	1	2	0
1	55	1	3	142	154	0	0	149	0	0.45	0	0	1	0
2	29	1	1	100	146	0	0	185	0	0.89	2	0	3	0
3	65	0	1	103	182	0	1	176	1	0.18	1	0	1	0
4	75	1	2	124	136	1	0	147	0	0.27	0	1	1	1

```
df['sex'].value_counts()
```

Show hidden output

```
df.describe()
```

Show hidden output

```
df.skew()
```

McAfee | WebAdvisor
Your download's being scanned.
We'll let you know if there's an issue.

	0
age	-0.008310
sex	-0.148630
cp	0.053778
trestbps	0.169872
chol	0.231097
fbs	1.952384
restecg	0.609639
thalach	-0.114602
exang	-0.044077
oldpeak	1.795567
slope	0.092311
ca	0.987652
thal	0.455757
target	0.000000

dtype: float64

```
df.kurt()
```

	0
age	-1.120297
sex	-1.981877
cp	-1.378974
trestbps	-0.294976
chol	-0.529107
fbs	1.815431
restecg	-0.679145
thalach	-0.556637
exang	-2.002065
oldpeak	4.050829
slope	-0.883052
ca	-0.032667
thal	-1.519083
target	-2.004012

dtype: float64

```
Q1 = df["chol"].quantile(0.25)
Q3 = df["chol"].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

df_clean = df[(df["chol"] >= lower) & (df["chol"] <= upper)]
```

```
# defining the featuree
X = df_clean[["age"]]
y = df_clean["chol"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
```

McAfee | WebAdvisor
Your download's being scanned.
We'll let you know if there's an issue.

```
)
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

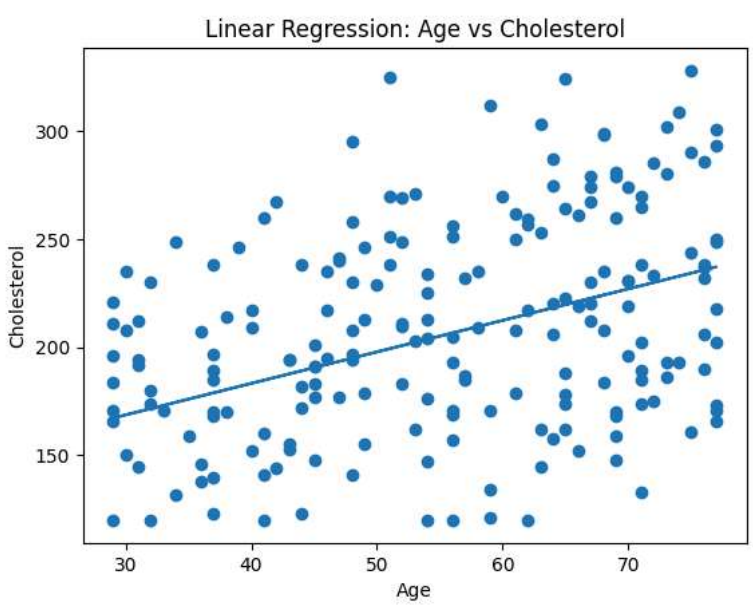
```
y_pred = model.predict(X_test)
```

```
import numpy as np
from sklearn.metrics import r2_score, mean_squared_error, mean_absolute_error
r2 = r2_score(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
mae = mean_absolute_error(y_test, y_pred)

n = len(y_test)
p = X_test.shape[1]

adjusted_r2 = 1 - (1 - r2) * (n - 1) / (n - p - 1)
```

```
print("----- Linear Regression Results -----")
print("Intercept:", model.intercept_)
print("Coefficient:", model.coef_[0])
print("R-square:", r2)
print("Adjusted R-square:", adjusted_r2)
print("MSE:", mse)
print("RMSE:", rmse)
print("MAE:", mae)
```

```
----- Linear Regression Results -----
Intercept: 124.9884157337792
Coefficient: 1.4571677574562842
R-square: 0.11343299343658819
Adjusted R-square: 0.10895538229232848
MSE: 2132.336188595486
RMSE: 46.17722586508945
MAE: 38.04622645596366
```

```
plt.figure()
plt.scatter(X_test, y_test)
plt.plot(X_test, y_pred)
plt.xlabel("Age")
plt.ylabel("Cholesterol")
plt.title("Linear Regression: Age vs Cholesterol")
plt.show()
```



McAfee | WebAdvisor
Your download's being scanned.
We'll let you know if there's an issue.

LOGISTIC REGRESSION

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score, f1_score
```

```
df = pd.read_csv("heart_disease_dataset.csv")
print(df.head())
```

```
   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  slope  \
0   39    1   1       102   123    0        1      194      0     0.31      1
1   55    1   3       142   154    0        0      149      0     0.45      0
2   29    1   1       100   146    0        0      185      0     0.89      2
3   65    0   1       103   182    0        1      176      1     0.18      1
4   75    1   2       124   136    1        0      147      0     0.27      0

   ca  thal  target
0   1     2       0
1   0     1       0
2   0     3       0
3   0     1       0
4   1     1       1
```

```
X = df[['age']]     # Independent variable
y = df['target']    # Dependent variable
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
▾ LogisticRegression
LogisticRegression(max_iter=1000)
```

```
y_pred = model.predict(X_test)
```

```
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)
```

```
Confusion Matrix:
 [[62 38]
 [30 70]]
```

```
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)
print("F1 Score:", f1)
```

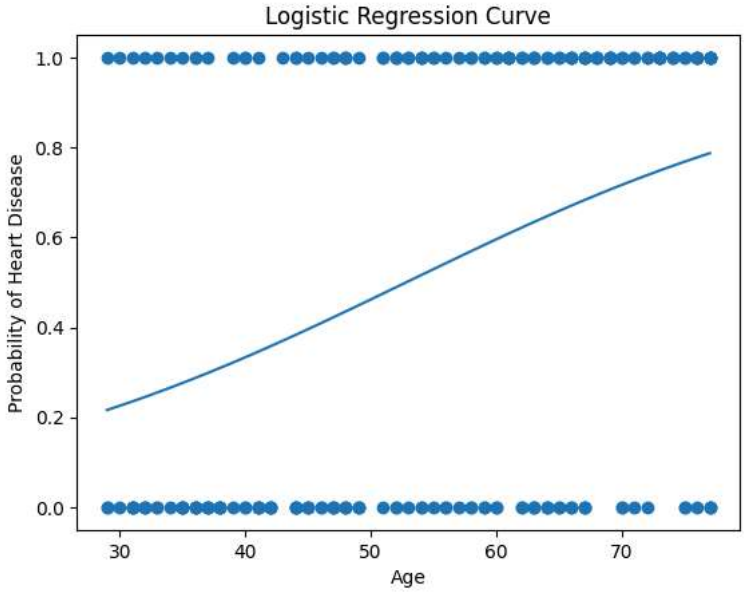
```
Accuracy: 0.66
Precision: 0.6481481481481481
Recall: 0.7
F1 Score: 0.6730769230769231
```

```
X_range = np.linspace(X.min(), X.max(), 300).reshape(-1,1)
y_prob = model.predict_proba(X_range)[:,1]

plt.figure()
plt.scatter(X_test, y_test)
plt.plot(X_range, y_prob)
plt.xlabel("Age")
```

McAfee | WebAdvisor
Your download's being scanned.
We'll let you know if there's an issue.

```
plt.ylabel("Probability of Heart Disease")
plt.title("Logistic Regression Curve")
plt.show()
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but
  warnings.warn(
```




```
X = df[['age', 'chol', 'trestbps', 'thalach', 'exang']]
y = df['target']


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)


model = LogisticRegression(max_iter=2000)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

```
cm = confusion_matrix(y_test, y_pred)

print("Confusion Matrix:")
print(cm)
```

```
Confusion Matrix:
[[70 30]
 [31 69]]
```

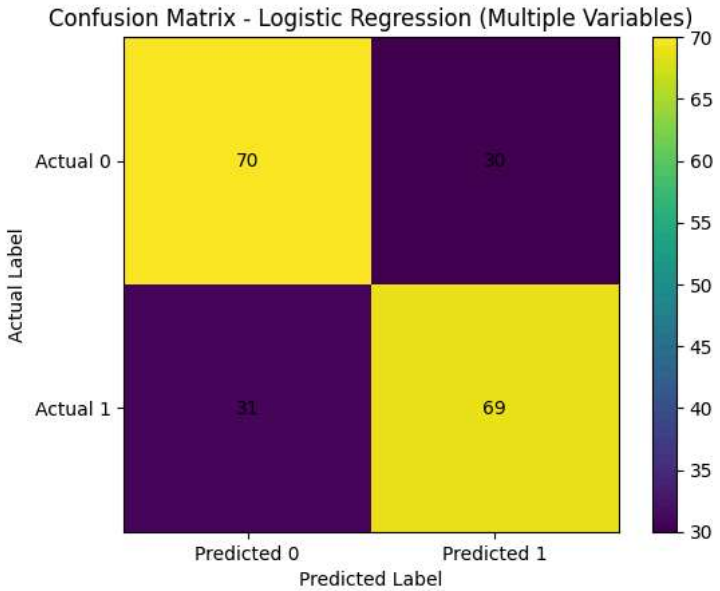
```
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("\nModel Performance:")
print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)
print("F1 Score:", f1)
```

```
Model Performance:
Accuracy: 0.695
Precision: 0.696969696969697
Recall: 0.69
F1 Score: 0.6934673366834171
```

```
plt.figure()
plt.imshow(cm)
plt.colorbar()
```

McAfee | WebAdvisor
Your download's being scanned.
We'll let you know if there's an issue.

```
plt.xticks([0, 1], ['Predicted 0', 'Predicted 1'])
plt.yticks([0, 1], ['Actual 0', 'Actual 1'])

for i in range(2):
    for j in range(2):
        plt.text(j, i, cm[i, j], ha='center', va='center')

plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.title("Confusion Matrix - Logistic Regression (Multiple Variables)")
plt.show()
```



MULTIPLE LINEAR REGRESSION

```
X = df[['age', 'trestbps', 'thalach', 'oldpeak']]
y = df['chol']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)
```

```
Intercept: 112.00394986634984
Coefficients: [1.56486069 0.03086782 0.02107155 1.55559805]
```

```
y_pred = model.predict(X_test)
```

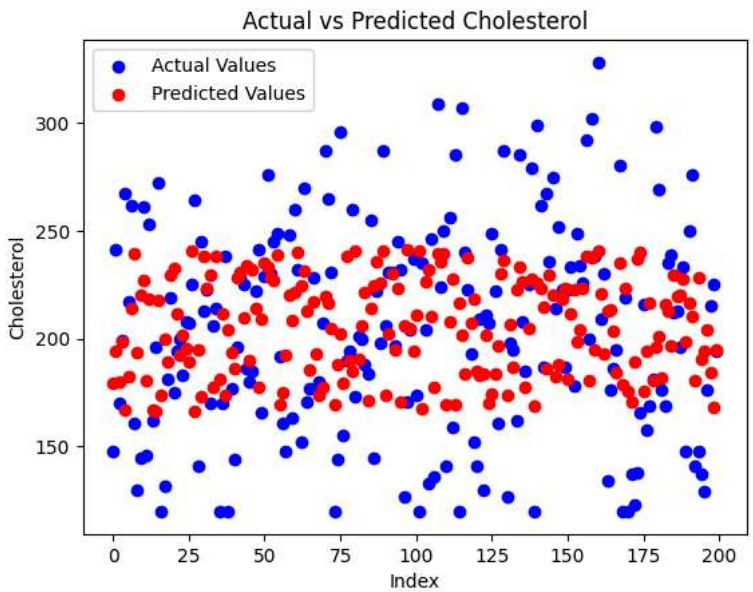
```
r2 = r2_score(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
mae = mean_absolute_error(y_test, y_pred)

n = len(y_test)
p = X_test.shape[1]

adjusted_r2 = 1 - (1 - r2) * (n - 1) / (n - p - 1)
```

McAfee | WebAdvisor
Your download's being scanned.
We'll let you know if there's an issue.

```
print("R-square:", r2)
print("Adjusted R-square:", adjusted_r2)
print("MSE:", mse)
print("RMSE:", rmse)
print("MAE:", mae)
```

```
R-square: 0.037093501846470534
Adjusted R-square: 0.017341573679218625
MSE: 2261.5471977520046
RMSE: 47.55572728654252
MAE: 39.37992997591566
```

```
plt.figure()

# Plot Actual values
plt.scatter(range(len(y_test)), y_test, color='blue', label='Actual Values')

# Plot Predicted values
plt.scatter(range(len(y_pred)), y_pred, color='red', label='Predicted Values')

plt.xlabel("Index")
plt.ylabel("Cholesterol")
plt.title("Actual vs Predicted Cholesterol")
plt.legend()

plt.show()
```



McAfee | WebAdvisor
Your download's being scanned.
We'll let you know if there's an issue.